

AGRIM JOSHI

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Computer Science undergraduate focused on applied Machine learning, with full-stack experience building end-to-end AI-Powered apps

EDUCATION

Simon Fraser University — Burnaby, BC

Expected 12/2027

Bachelor of Applied Science, Computing Science

PROFESSIONAL EXPERIENCE

Special Projects Coordinator — Software Developer

Jan 2026 — Present

SFU Beedie School of Business — Burnaby, BC

Co-op: Jan–Apr 2026 | Contractor (Part-Time): Sept 2026– Present

- Developed a **full-stack degree planning web application** that helps SFU Beedie students choose courses, check degree requirements, and plan future semesters.
- Built the **frontend and backend using PHP** and designed a **PostgreSQL database** to manage 181 courses, degree requirements, concentrations, prerequisites, and student selections.
- Developed **REST APIs** to connect the web application with other systems and support application integrations, and created tools to summarize expected course demand, helping administrators improve semester planning.

Programming Instructor

Sept 2025 — Dec 2025

Wize Computing Academy — Vancouver, BC

- Designed and delivered interactive lessons on Python fundamentals, covering algorithms, problem-solving, and debugging.
- Introduced 3D modelling tools to build creativity and spatial reasoning, preparing students for engineering and design applications.

TECHNICAL PROJECTS

ShadowAthlete - Personalized AI-powered Sports Analytics and Training Application

Oct 2026

Python, FastAPI, React Native, PostgreSQL, Redis, MMpose, YOLOv8, OpenCV, PyTorch, REST APIs, JWT, Postman, Docker

- Built a **full-stack AI** sports training platform that analyzes athlete videos across 4 different sports and generates an evolving digital avatar which evolves as the user improves.
- Developed a computer vision pipeline using **MMpose, YOLOv8 and PyTorch** that extracts 33 body landmarks per frame and converts that data into multiple performance metrics, processing a 30-second clip in 10 seconds.
- Engineered a **FastAPI backend with 15+ REST modules, PostgreSQL, Redis caching, JWT authentication, and Postman API testing**, containerized with **Docker** and integrated with the **React Native** frontend

Re-Habit — AI Motion-Tracking Web-App for Physiotherapy

Oct 2025

Python, Flask, MediaPipe, OpenCV, Pose Estimation

- Built a **camera-based web application** that tracks a patient's body movements in real time and guides them through stroke-recovery exercises.
- Used **MediaPipe** and **OpenCV** to track joints, count exercise repetitions, and detect incorrect movements while reducing false counts.
- Created a dashboard that measures movement symmetry, posture, and walking patterns and turns the results into simple progress summaries.

Medical VQA Analysis - Vision Language Model for Clinical Procedures

Aug 2025

Python, PyTorch, Hugging Face Transformers, OpenCV, Computer Vision, CLIP, ViLT

- Built and evaluated a medical Visual Question Answering pipeline across 2,000+ image-question pairs spanning 13+ clinical question categories, preparing structured image, question, and answer data for model training and evaluation.
- Performed annotation and dataset sanitization by manually reviewing medical images and question-answer labels, correcting mislabeled samples, standardizing inconsistent annotations, and flagging ambiguous cases to improve dataset quality.
- Fine-tuned **CLIP and ViLT** for medical visual question answering, improving accuracy by 35% over zero-shot baselines.

SKILLS

Languages/Database : C/C++, Python, JavaScript, HTML, CSS, SQL, PHP, PostgreSQL, MongoDB, R, MATLAB

Frameworks/ Libraries : React, React Native, Node.js, Flask, Pandas, NumPy

AI/ML : TensorFlow, PyTorch, OpenCV, Pose Estimation, Scikit-Learn, Data Annotation, YOLO, Mediapipe

Web/API & Security : REST APIs, FastAPI, OAuth 2.0, JWT, CORS, WebSockets, JSON, API Authentication

Tools : Git, GitHub, Docker, Postman, Redis, Linux, CLI, Neovim

ACHIEVEMENTS & CERTIFICATES

- Placed Top 3 at the Stormhacks hackathon, for the project Re-Habit, out of 250+ teams and 1000+ students.
- Data Analyst Certificate, Coursera (Oct 2024) | Data Structures & Algorithms, ITI Informatics (Aug–Dec 2023).